

Project Name: DriveShare - Car rental platform
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Youtube link: <https://youtu.be/Zi5gZfTBk24>

Github link: <https://github.com/Michael-V-21/CIS476TermProjectDriveShare.git>

Design Patterns:

Singleton Pattern - Makes sure 1 instance of a class exists through the whole application.

- Role: Manage the user's session securely.
- Implemented in: SessionManager

Observer Pattern - Allows objects to be notified automatically when another object changes.

- Role: Notify renters if their watched cars meet their expectations.
- Implemented in: CarSubject, CarObserver, RenterObserver, Car

Mediator Pattern - Centralizes communication between multiple objects (reduces dependencies).

- Role: Manage communication between different UI components, creating a cohesive and user-friendly interface.
- Implemented in: UIMediator, DriveShareMediator, DashboardFrame

Builder Pattern - Constructs objects (with optional attributes) one step at a time.

- Role: Construct car listing objects with optional/variable attributes cleanly.
- Implemented in: CarBuilder, Car, CreateCarFrame

Proxy Pattern - Placeholder to control access to another object.

- Role: Simulate secure interaction between DriveShare and the payment system.
- Implemented in: PaymentService, PaymentProxy, RealPaymentService, PaymentFrame

Chain of Responsibility Pattern - Passes a request along a chain of handlers until its processed.

- Role: Create a secure process, using three security questions to build the chain, for recovering a forgotten password.
- Implemented in: SecurityQuestionHandler, Question1Handler, Question2Handler, Question3Handler, RecoverPasswordFrame

Session ==> SessionManager

Observer ==> CarSubject (Subject), CarObserver (Observer), RenterObserver (Concrete Observer)

Mediator ==> UIMediator (Mediator), DriveShareMediator (Concrete Mediator)

Builder ==> CarBuilder (Builder), Car (Product)

Proxy ==> PaymentService (Subject), PaymentProxy (Proxy), RealPaymentService (Real Subject)

Chain of Responsibility ==> SecurityQuestionHandler (Handler), Question1Handler, Question2Handler, Question3Handler

Database Schema and Descriptions

- I did not use a database in my implementation of this project, as it wasn't required in the rubric. I used in-memory data storage to manage all the data. Below I have a simple structure that would be the core of an actual database.

Users:

- email (primary identifier)
- password
- security questions and answers

Cars:

- carId (primary identifier)
- ownerEmail
- model, year, mileage, location, pricePerDay
- availability (availableFrom, availableTo)

Bookings:

- carId
- renterEmail
- startDate, endDate
- totalCost
- paid status

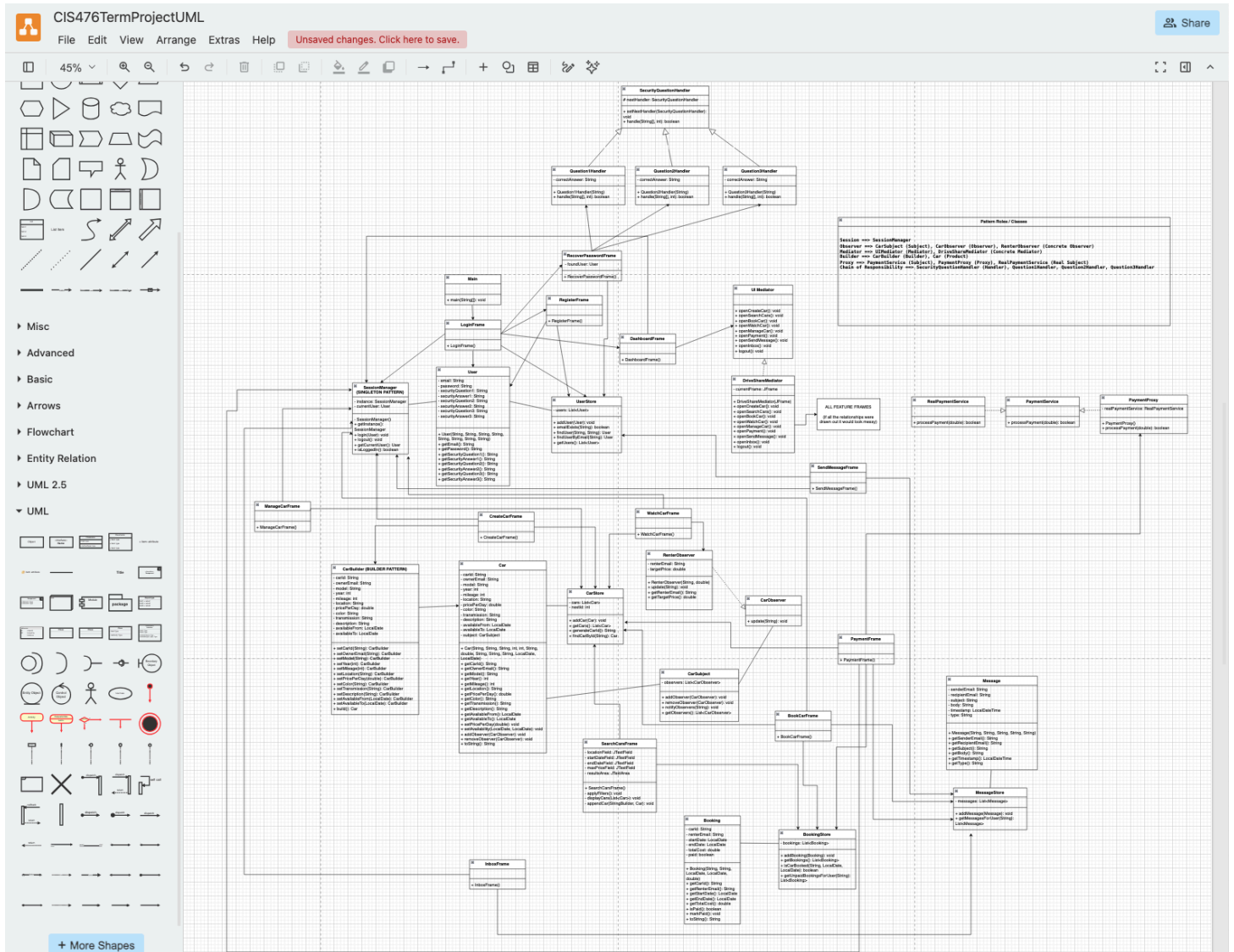
Messages:

- senderEmail
- recipientEmail
- subject, body
- timestamp
- type (user message or system notification)

Task Summary

- Project was completed individually, so I implemented all tasks
- User registration and login with session management
- Car listing creation and management
- Car search and booking system
- Watch feature with notification system
- Messaging system between users
- Payment simulation using proxy pattern
- Password recovery using 3 security questions
- Implementation of all design patterns (Singleton, Observer, Mediator, Builder, Proxy, Chain)
- UML Class diagram
- UI using Java swing

UML Diagram



*** This UML class diagram shows the system structure of my Driveshare application, with all the classes and their relationships to one another. It also highlights the implementation of the design patterns and their corresponding classes.

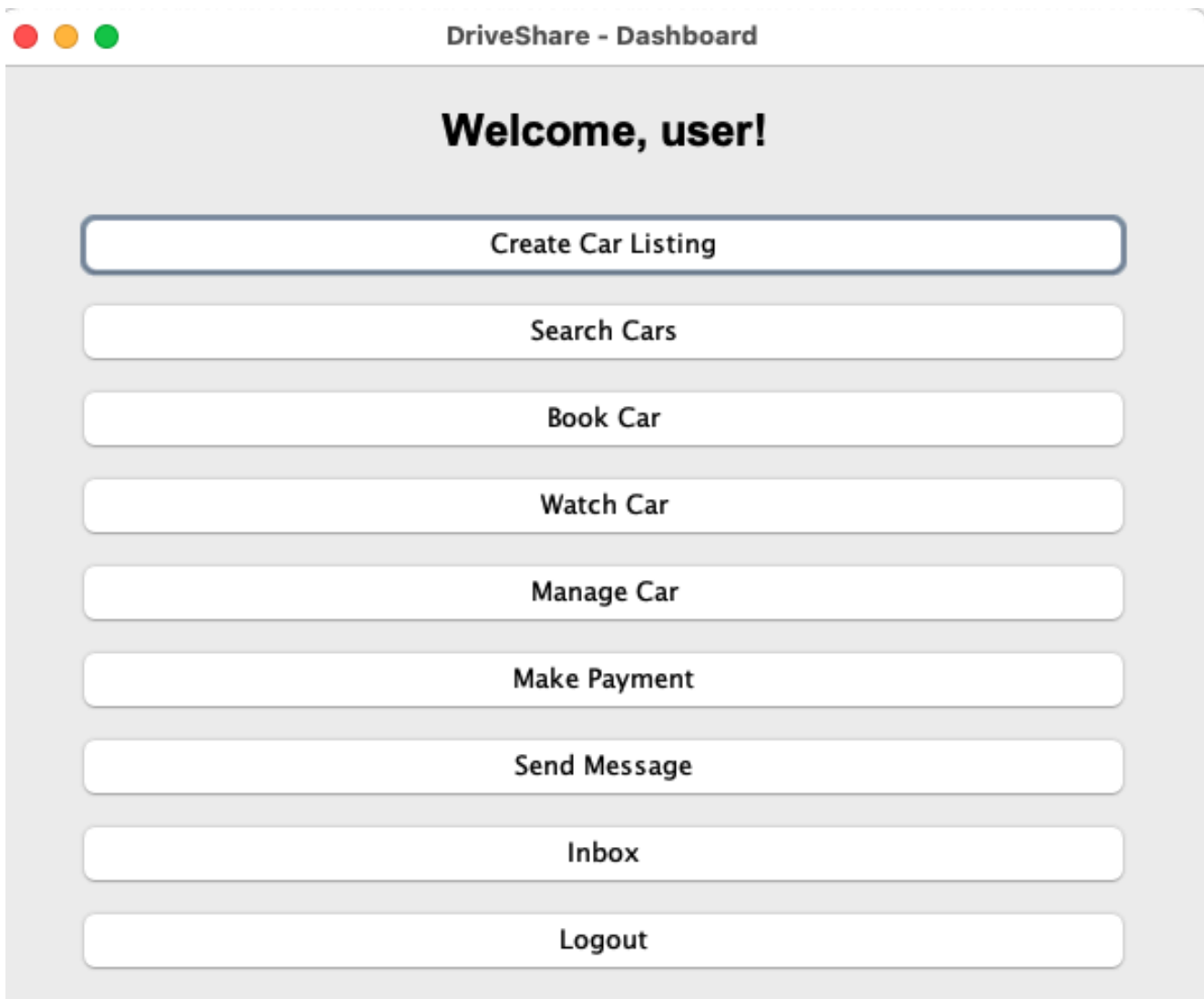
The official UML class diagram is included in the zip file at a better picture quality.

References

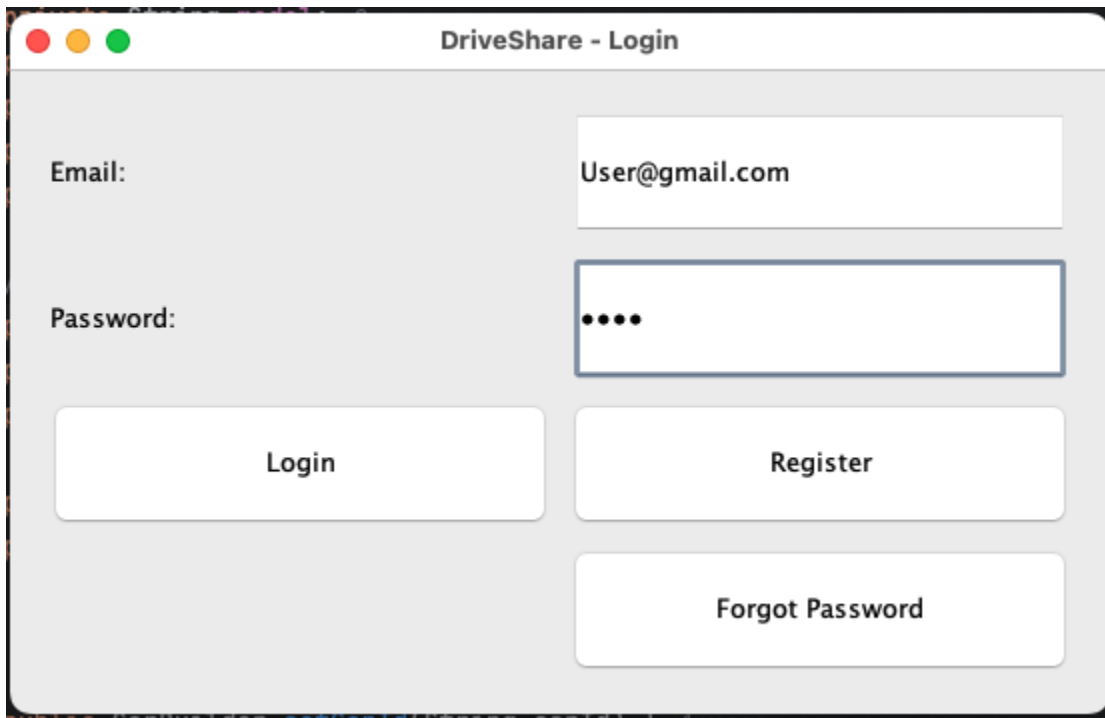
- [Turo.com](#)
- Java Swing documentation
- CIS 476 Course Lecture Slides + textbook (for patterns)

UI Screenshots:

Dashboard - Displays the main menu where users can navigate to all app features.



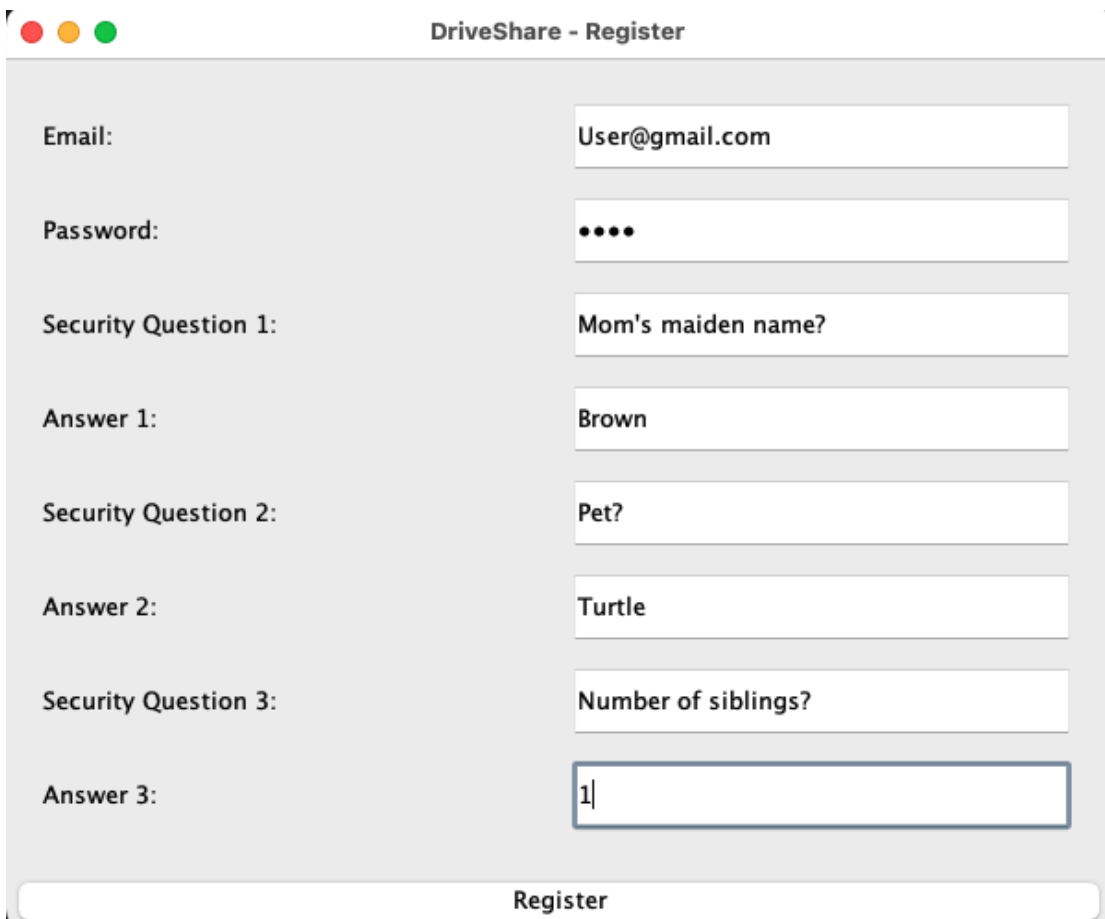
Login Screen - Allows registered users to log into the DriveShare system.



The image shows a login window titled "DriveShare - Login". It features a light gray background with three input fields on the right: "Email:" with the value "User@gmail.com", "Password:" with four dots, and a "Forgot Password" link. Below the password field are two buttons: "Login" and "Register".

Email:	User@gmail.com
Password:	••••
Forgot Password	
Login	Register

Registration Screen - Allows new users to create an account by inputting credentials and security questions.



The image shows a registration window titled "DriveShare - Register". It features a light gray background with six input fields on the right: "Email:" with the value "User@gmail.com", "Password:" with four dots, "Security Question 1:" with the value "Mom's maiden name?", "Answer 1:" with the value "Brown", "Security Question 2:" with the value "Pet?", "Answer 2:" with the value "Turtle", "Security Question 3:" with the value "Number of siblings?", and "Answer 3:" with the value "1". A "Register" button is at the bottom.

Email:	User@gmail.com
Password:	••••
Security Question 1:	Mom's maiden name?
Answer 1:	Brown
Security Question 2:	Pet?
Answer 2:	Turtle
Security Question 3:	Number of siblings?
Answer 3:	1
Register	

Create Car Screen - Allows car owners to list vehicles with car details, availability, and price.

DriveShare - Create Car Listing

Car Model:

Honda Civic

Year:

2018

Mileage:

200000

Pickup Location:

Dearborn

Price Per Day:

50

Available From (YYYY-MM-DD):

2026-04-10

Available To (YYYY-MM-DD):

2026-04-30

Color:

Black

Transmission:

Manual

Description:

Fast

Create Listing

Search Car Screen - Allows renters to search for listed cars based on certain filters.

DriveShare - Search Cars

Location:

Start Date (YYYY-MM-DD):

End Date (YYYY-MM-DD):

Max Price Per Day:

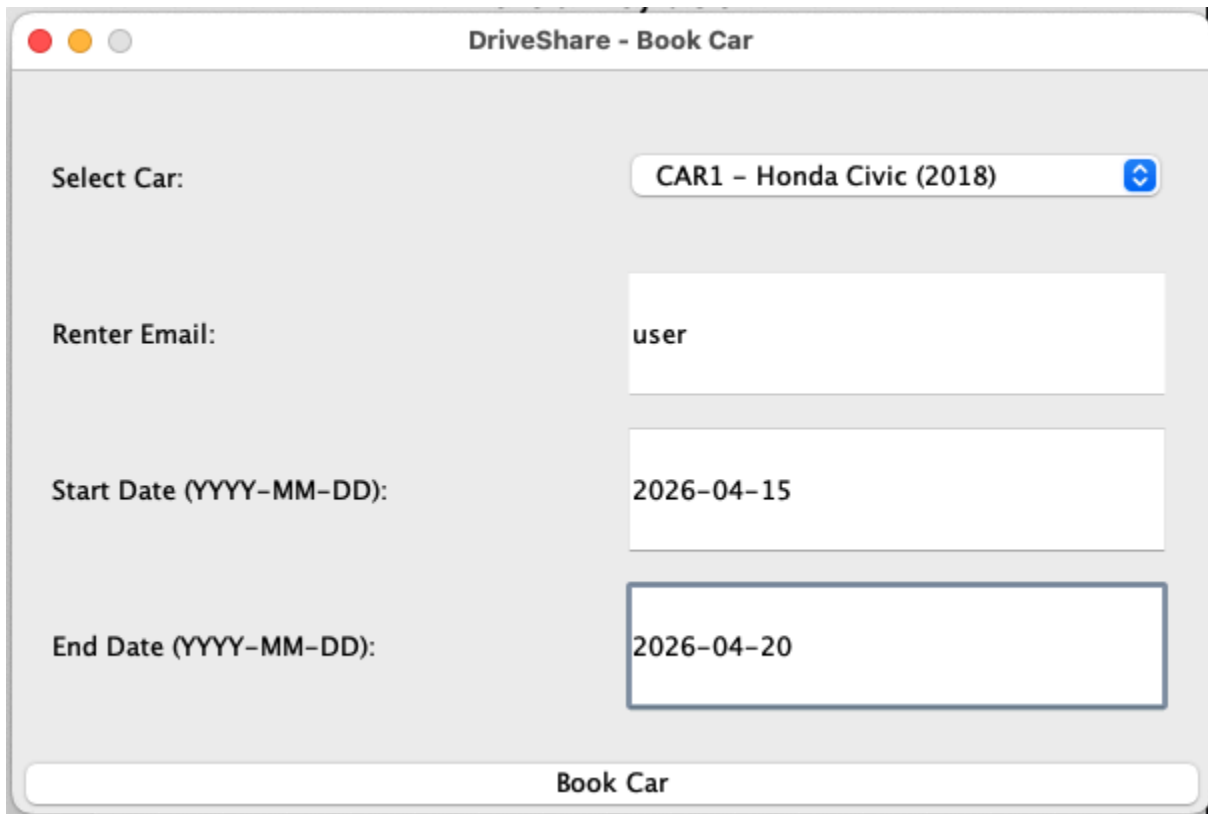
Apply Filters

Show All Cars

Car ID: CAR1
Owner: owner
Model: Honda Civic
Year: 2018
Mileage: 200000
Location: Dearborn
Price Per Day: \$50.0
Available From: 2026-04-10
Available To: 2026-04-30
Color: Black
Transmission: Manual
Description: Fast

Car ID: CAR2
Owner: owner
Model: Cadillac ATS
Year: 2020
Mileage: 100000
Location: Detroit
Price Per Day: \$100.0
Available From: 2026-05-01
Available To: 2026-05-30

Book Car Screen - Allow users to book cars for certain dates.



A screenshot of a web application window titled "DriveShare - Book Car". The window has a light gray background and a white border. It contains four input fields and a button. The first field is a dropdown menu labeled "Select Car:" with the text "CAR1 - Honda Civic (2018)" and a blue arrow icon. The second field is a text input labeled "Renter Email:" with the text "user". The third field is a text input labeled "Start Date (YYYY-MM-DD):" with the text "2026-04-15". The fourth field is a text input labeled "End Date (YYYY-MM-DD):" with the text "2026-04-20". At the bottom of the form is a white button labeled "Book Car".

DriveShare - Book Car

Select Car: CAR1 - Honda Civic (2018)

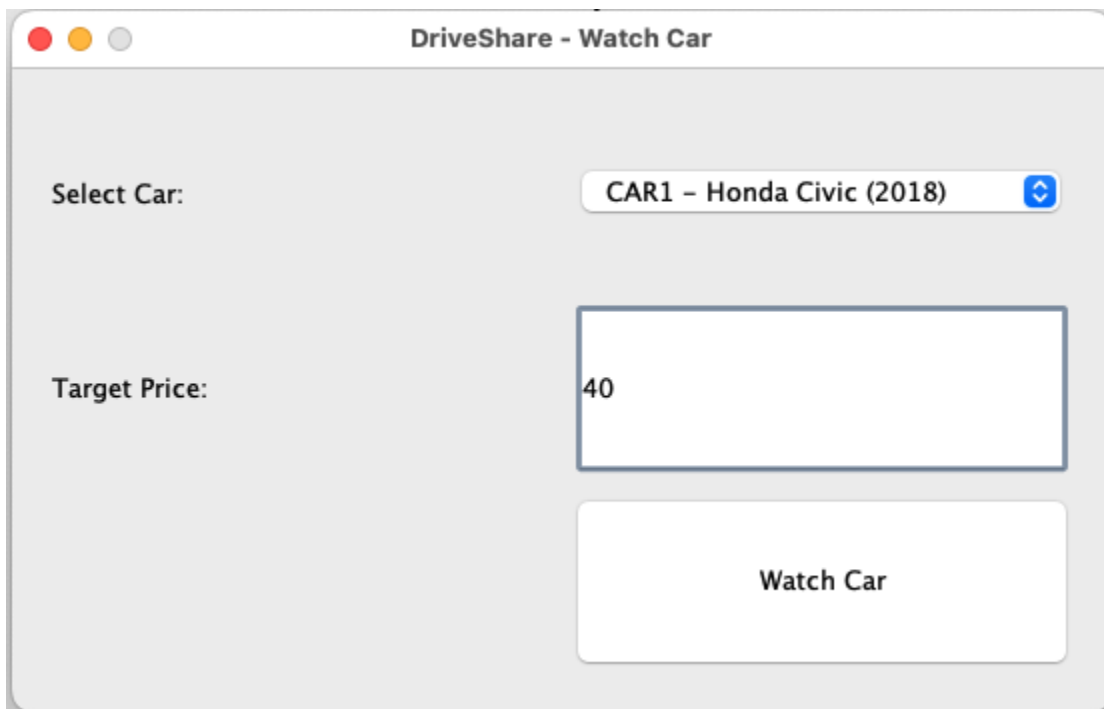
Renter Email: user

Start Date (YYYY-MM-DD): 2026-04-15

End Date (YYYY-MM-DD): 2026-04-20

Book Car

Watch Car Screen - Allow users to watch cars and get notifications for price updates.



A screenshot of a web application window titled "DriveShare - Watch Car". The window has a light gray background and a white border. It contains two input fields and a button. The first field is a dropdown menu labeled "Select Car:" with the text "CAR1 - Honda Civic (2018)" and a blue arrow icon. The second field is a text input labeled "Target Price:" with the text "40". At the bottom of the form is a white button labeled "Watch Car".

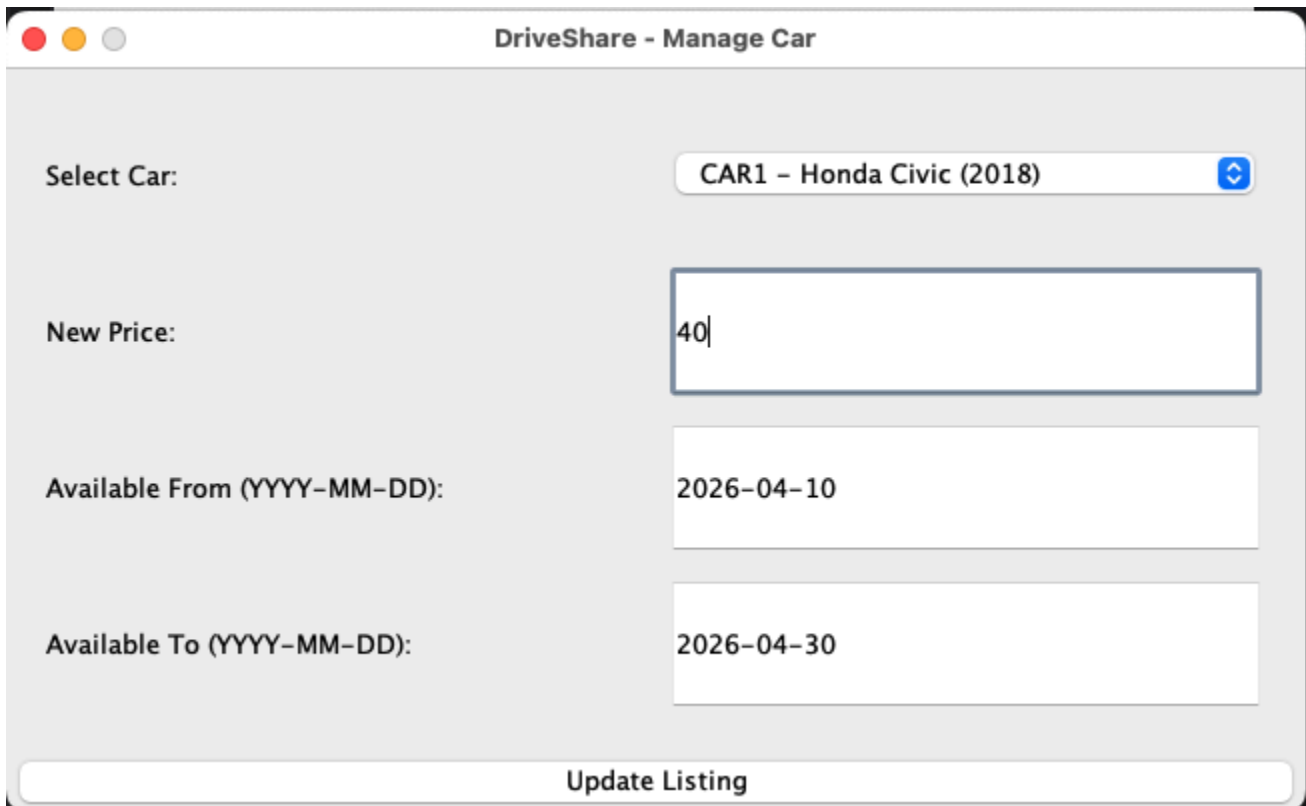
DriveShare - Watch Car

Select Car: CAR1 - Honda Civic (2018)

Target Price: 40

Watch Car

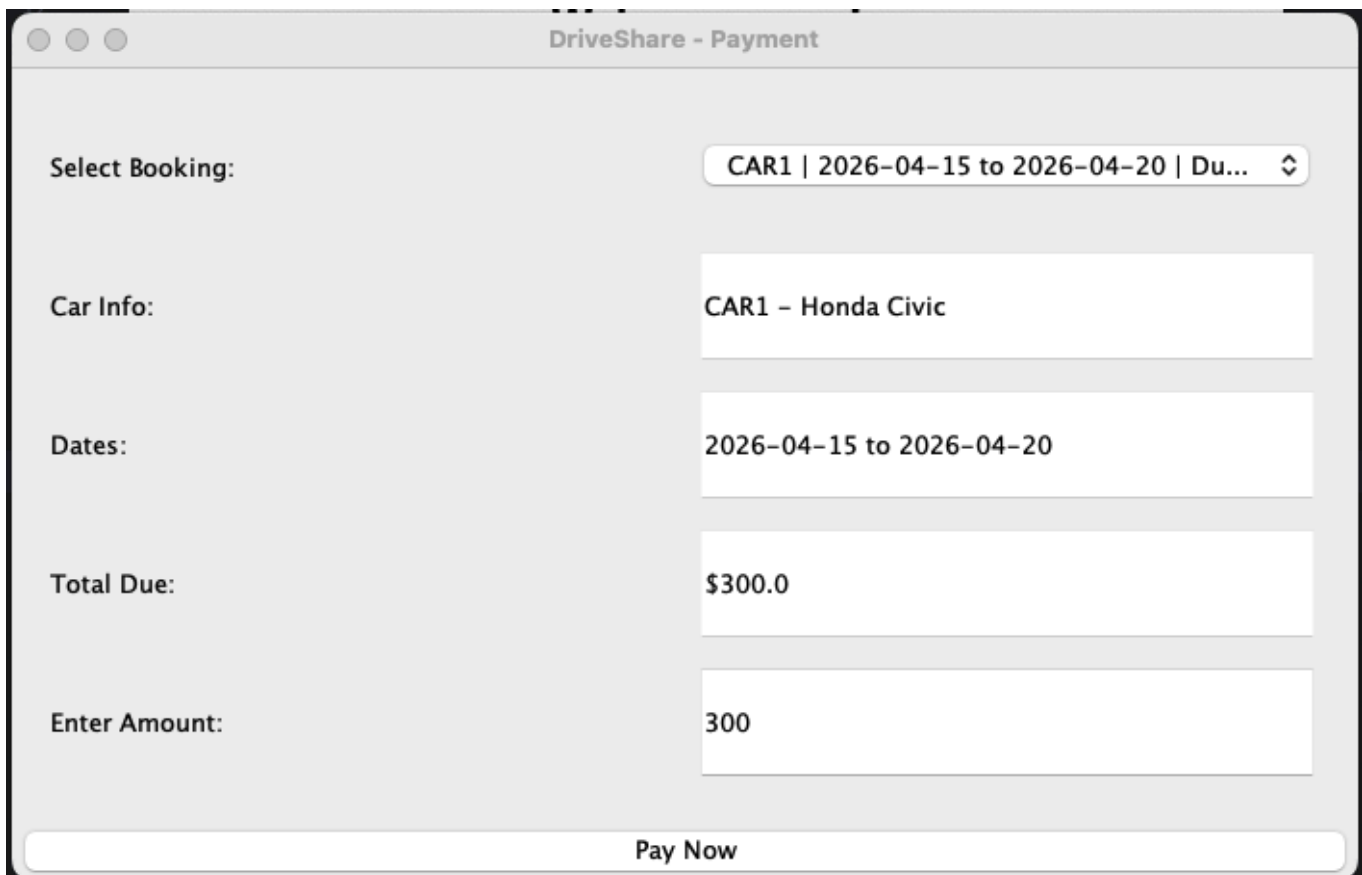
Manage Car Screen - Allow car owners only to manage their car, updating price and availability dates.



A screenshot of a web application window titled "DriveShare - Manage Car". The window has a light gray background and rounded corners. It contains several form fields for managing a car listing. At the top, there's a "Select Car:" label followed by a dropdown menu showing "CAR1 - Honda Civic (2018)" with a blue arrow icon. Below this is a "New Price:" label followed by a text input field containing "40". Next is an "Available From (YYYY-MM-DD):" label followed by a text input field containing "2026-04-10". Below that is an "Available To (YYYY-MM-DD):" label followed by a text input field containing "2026-04-30". At the bottom, there's a wide button labeled "Update Listing".

Select Car:	CAR1 - Honda Civic (2018)
New Price:	40
Available From (YYYY-MM-DD):	2026-04-10
Available To (YYYY-MM-DD):	2026-04-30
Update Listing	

Payment Screen - Simulates payment processing for bookings.



A screenshot of a web application window titled "DriveShare - Payment". The window has a light gray background and rounded corners. It contains several form fields for processing a payment. At the top, there's a "Select Booking:" label followed by a dropdown menu showing "CAR1 | 2026-04-15 to 2026-04-20 | Du..." with a blue arrow icon. Below this is a "Car Info:" label followed by a text input field containing "CAR1 - Honda Civic". Next is a "Dates:" label followed by a text input field containing "2026-04-15 to 2026-04-20". Below that is a "Total Due:" label followed by a text input field containing "\$300.0". At the bottom, there's an "Enter Amount:" label followed by a text input field containing "300". At the very bottom, there's a wide button labeled "Pay Now".

Select Booking:	CAR1 2026-04-15 to 2026-04-20 Du...
Car Info:	CAR1 - Honda Civic
Dates:	2026-04-15 to 2026-04-20
Total Due:	\$300.0
Enter Amount:	300
Pay Now	

Send Message Screen - Allows users to send messages to other users.

DriveShare - Send Message

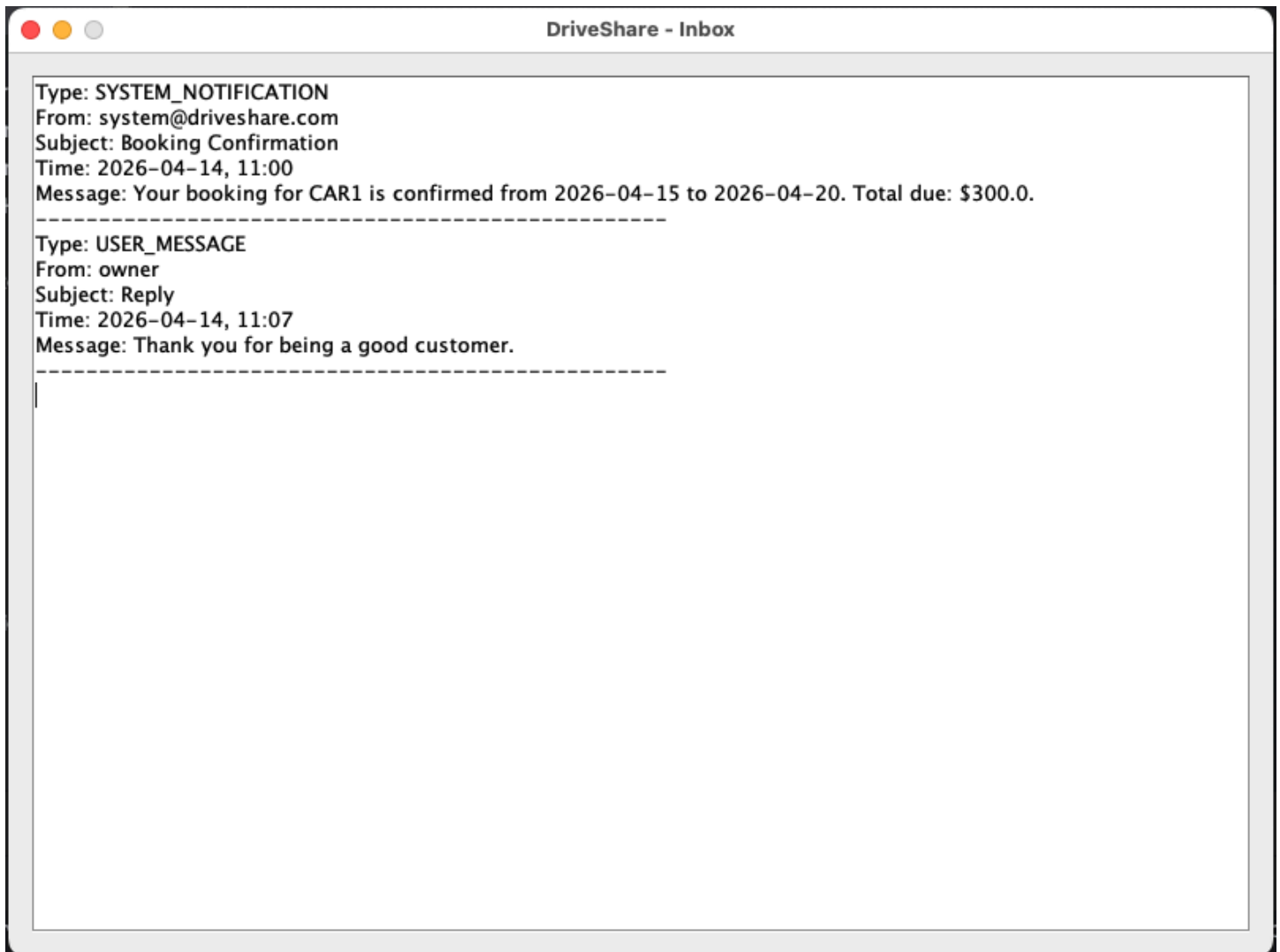
To: User@gmail.com

Subject: Car rental Honda Civic

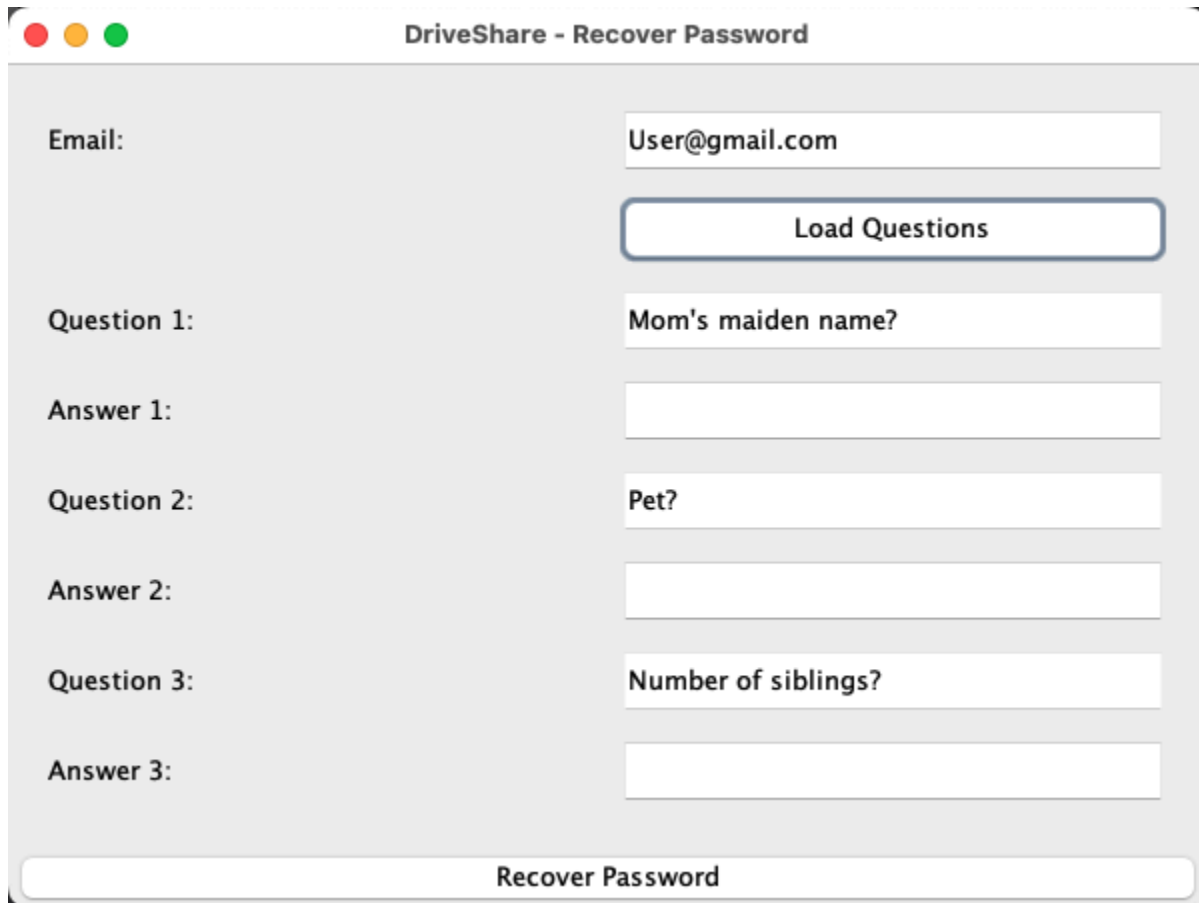
Message: I would like to thank you for the car. |

Send Message

Inbox Screen - Allows users to see their received messages and notifications.



Recover Password Screen - Allows users to recover their password using email and security questions.



The image shows a web form titled "DriveShare - Recover Password". It features a light gray background with white input fields and a blue "Load Questions" button. The form is organized into sections for email, three security questions, and their corresponding answers. At the bottom, there is a wide "Recover Password" button.

Field	Value
Email:	User@gmail.com
Load Questions	
Question 1:	Mom's maiden name?
Answer 1:	
Question 2:	Pet?
Answer 2:	
Question 3:	Number of siblings?
Answer 3:	
Recover Password	